

Chapter 6 — Solving the linear system

Conditioning of finite element matrices, direct and iterative solvers

MTH3340 · Numerical Methods for PDEs

1 Where the cost is

Every chapter so far ended in the same place:

$$Au = f, \quad A \in \mathbb{R}^{n \times n}, \quad f \in \mathbb{R}^n, \quad u_h(\mathbf{x}) = \sum_{a=1}^n u^a \phi^a(\mathbf{x}).$$

- A is square and, for a well-posed problem, **non-singular**: unique solution;
- u_h does not depend on the basis of V_h ; A, f, u do;
- n grows with refinement: $n \sim h^{-d}$. For $h = 10^{-2}$: 10^2 unknowns in 1D, 10^4 in 2D, 10^6 in 3D.

In a finite element code, the solution of $Au = f$ dominates the computational cost.

Today: (1) how *hard* are finite element matrices? — conditioning, §6.3; (2) solve exactly: **direct methods**, §6.4; (3) solve approximately: **iterative methods** and preconditioning, §6.5.

2 Eigenvalues and the condition number

$Au = \lambda u, u \neq 0$. **Spectrum** $\sigma(A) \subset \mathbb{C}$; **spectral radius** $\rho(A) \doteq \max_{\lambda \in \sigma(A)} |\lambda|$; **kernel** $\text{Ker}(A) \doteq \{u : Au = 0\}$ (non-trivial iff A singular).

Induced **matrix norm** and **condition number**, for a vector norm $\|\cdot\|_*$ (e.g. $\|u\|_2$, or $\|u\|_A \doteq \sqrt{u^T A u}$ for A s.p.d.):

$$\|B\|_* \doteq \max_{u \neq 0} \frac{\|Bu\|_*}{\|u\|_*}, \quad \kappa_*(A) \doteq \|A\|_* \|A^{-1}\|_* \geq 1.$$

$\kappa = \infty$ if A is singular. κ measures how *hard* the system is: it controls the amplification of **rounding errors** and, today, the speed of **iterative convergence**.

Q: What is $\kappa(A)$ for $A = c \mathbf{I}$, $c \in \mathbb{R}$, $c \neq 0$?

3 The condition number of an s.p.d. matrix

A symmetric positive definite ($u^T A u > 0$ for $u \neq 0$): real positive eigenvalues $\lambda_{\min} \leq \dots \leq \lambda_{\max}$ and an orthonormal eigenbasis $\{v_i\}$, $v_i^T v_j = \delta_{ij}$.

Expand $u = \sum_i \alpha_i v_i$, so that $Au = \sum_i \alpha_i \lambda_i v_i$ and $\|u\|_2^2 = \sum_i \alpha_i^2$:

$$\|A\|_2 = \max_{u \neq 0} \frac{\|\sum_i \alpha_i \lambda_i v_i\|_2}{\|\sum_i \alpha_i v_i\|_2} = \max_{\alpha \neq 0} \sqrt{\frac{\sum_i \alpha_i^2 \lambda_i^2}{\sum_i \alpha_i^2}} = \lambda_{\max},$$

$$A^{-1} v_i = \frac{1}{\lambda_i} v_i \Rightarrow \|A^{-1}\|_2 = \frac{1}{\lambda_{\min}}.$$

Hence

$$\kappa_2(A) = \frac{\lambda_{\max}}{\lambda_{\min}} :$$

well conditioned if $\kappa_2 \approx 1$; ill conditioned as $\lambda_{\min} \rightarrow 0$ or $\lambda_{\max} \rightarrow \infty$.

4 Rayleigh quotients

Courant–Fischer (min-max): for A symmetric, the extreme eigenvalues are extreme values of the **Rayleigh quotient**

$$R_A(u) \doteq \frac{(Au, u)}{(u, u)}, \quad \lambda_{\max} = \max_{u \neq 0} R_A(u), \quad \lambda_{\min} = \min_{u \neq 0} R_A(u).$$

So: bounding $u^T Au$ above and below by $c u^T u$ bounds the spectrum.

Lemma (Generalised eigenvalue problems — notes, Lemma 6.2.1)

Let A, B be s.p.d. The n eigenvalues of

$$Au = \lambda Bu, \quad B^{-1}Au = \lambda u, \quad B^{-\frac{1}{2}}AB^{-\frac{1}{2}}u = \lambda u$$

coincide; hence $\kappa(B^{-1}A) = \kappa(B^{-\frac{1}{2}}AB^{-\frac{1}{2}})$, and

$$\lambda_{\min} = \min_{u \neq 0} \frac{u^T Au}{u^T Bu}, \quad \lambda_{\max} = \max_{u \neq 0} \frac{u^T Au}{u^T Bu}.$$

5 Finite element matrices are inner products

Reaction and Laplacian terms on $V_h = \text{span}\{\phi^1, \dots, \phi^n\}$:

$$M_{ab} \doteq \int_{\Omega} \phi^a \phi^b, \quad L_{ab} \doteq \int_{\Omega} \nabla \phi^a \cdot \nabla \phi^b, \quad a, b \in \{1, \dots, n\}.$$

With $u_h = \sum_a \phi^a u^a$ and $v_h = \sum_b \phi^b v^b$, and taking $v = u$ on the right:

$$u^T M v = \int_{\Omega} u_h v_h, \quad u^T M u = \|u_h\|_{L^2}^2,$$

$$u^T L v = \int_{\Omega} \nabla u_h \cdot \nabla v_h, \quad u^T L u = \|\nabla u_h\|_{L^2}^2 = |u_h|_{H^1}^2,$$

$$u^T (M + L) v = \int_{\Omega} (u_h v_h + \nabla u_h \cdot \nabla v_h), \quad u^T (M + L) u = \|u_h\|_{H^1}^2 :$$

the L^2 , H_0^1 and H^1 inner products, and the corresponding squared norms.

The dictionary: a statement about the matrix $\leftrightarrow$ a statement about the finite element function. Rayleigh quotients of M and L compare $\|u_h\|_{L^2}$, $|u_h|_{H^1}$ with the *Euclidean* norm of the DOF vector.

6 The mass matrix is well conditioned

Theorem (Spectral bounds for the mass matrix)

On a shape-regular mesh of size h in $\mathbb{R}^d$,

$$c h^d \leq \frac{\langle Mv, v \rangle}{\langle v, v \rangle} \leq C h^d \quad \forall v \in \mathbb{R}^n \setminus \{0\},$$

i.e. $\lambda_{\min}(M)$ and $\lambda_{\max}(M)$ are both $\sim h^d$.

Split into cells and pull back to $\hat{K}$; $gl(a, K)$ is the global-to-local index:

$$\int_{\Omega} u_h^2 = \sum_{a,b} u^a u^b \sum_{K \in \mathcal{T}_h} \int_{\hat{K}} \hat{\phi}^{gl(a,K)} \hat{\phi}^{gl(b,K)} |J_K| d\hat{K}, \quad |J_K| = C h^d.$$

What is left is the **reference mass matrix** $\hat{M}_{ab} = \int_{\hat{K}} \hat{\phi}^a \hat{\phi}^b$, $a, b \leq \hat{n}$: s.p.d., **independent of h** . Each node has a bounded number of neighbours, so

$$C h^d \lambda_{\min}(\hat{M}) \langle u, u \rangle \leq \int_{\Omega} u_h^2 \leq C h^d \lambda_{\max}(\hat{M}) \langle u, u \rangle. \quad \square$$

$\kappa(M) = O(1)$: refining the mesh does not hurt.

7 The stiffness matrix is not

Theorem (Spectral bounds for the Laplacian matrix)

$$c h^d \leq \frac{\langle Lv, v \rangle}{\langle v, v \rangle} \leq C h^{d-2} \quad \forall v \in \mathbb{R}^n \setminus \{0\},$$

i.e. $\lambda_{\min}(L) \geq c h^d$ and $\lambda_{\max}(L) \leq C h^{d-2}$.

Upper: the inverse inequality (Week 8), then the mass bound,

$$\langle Lu, u \rangle = \|\nabla u_h\|^2 \leq C h^{-2} \|u_h\|^2 \leq C h^{d-2} \langle u, u \rangle;$$

lower: the Poincaré inequality, then the mass bound,

$$\langle Lu, u \rangle = \|\nabla u_h\|^2 \geq C \|u_h\|^2 \geq C h^d \langle u, u \rangle. \quad \square$$

$\kappa(L) = O(h^{-2})$: halve h , four times worse. Refinement makes the system *harder*, not only bigger.

8 The 1D numbers

Linear elements on a uniform mesh of $(0, 1)$, $h = 1/(n + 1)$; both matrices are tridiagonal and their eigenvalues are known in closed form.

Mass: $M = \frac{h}{6} \text{tridiag}(1, 4, 1)$, eigenvalues $\frac{h}{3}(2 + \cos \frac{k\pi}{n+1}) \in (\frac{h}{3}, h)$, so $\kappa(M) < 3$ for every h .

Stiffness: $L = \frac{1}{h} \text{tridiag}(-1, 2, -1)$, eigenvalues $\lambda_k = \frac{2}{h}(1 - \cos(k\pi h))$, so $\lambda_{\min} \approx \pi^2 h$ and $\lambda_{\max} \approx 4/h$: $\kappa(L) \approx 4/(\pi^2 h^2)$.

h	$\lambda_{\min}$	$\lambda_{\max}$	$\kappa(L)$
10^{-1}	0.98	39	40
10^{-2}	0.099	400	$4.1 \cdot 10^3$
10^{-3}	0.0099	4000	$4.1 \cdot 10^5$

9 Direct methods: Gaussian elimination

Robust, for *any* non-singular matrix. Eliminate column by column, $A \doteq A^{(0)}$:

$$\begin{bmatrix} a_{11}^{(0)} & a_{12}^{(0)} & \cdots & a_{1n}^{(0)} \\ a_{21}^{(0)} & a_{22}^{(0)} & \cdots & a_{2n}^{(0)} \\ \vdots & \vdots & & \vdots \\ a_{n1}^{(0)} & a_{n2}^{(0)} & \cdots & a_{nn}^{(0)} \end{bmatrix} \rightarrow \begin{bmatrix} a_{11}^{(0)} & a_{12}^{(0)} & \cdots & a_{1n}^{(0)} \\ 0 & a_{22}^{(1)} & \cdots & a_{2n}^{(1)} \\ \vdots & \vdots & & \vdots \\ 0 & a_{n2}^{(1)} & \cdots & a_{nn}^{(1)} \end{bmatrix} \rightarrow \cdots \rightarrow \begin{bmatrix} a_{11}^{(0)} & a_{12}^{(0)} & \cdots & a_{1n}^{(0)} \\ 0 & a_{22}^{(1)} & \cdots & a_{2n}^{(1)} \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn}^{(n-1)} \end{bmatrix} = U.$$

Step 1 replaces row i by $A(i, :) - l_{i1} A(1, :)$ with the **multipliers** $l_{i1} \doteq a_{i1}^{(0)} / a_{11}^{(0)}$, $i = 2, \dots, n$. In matrix form, $A^{(1)} = L_{(1)}^{-1} A$:

$$L_{(1)}^{-1} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ -l_{21} & 1 & & \\ \vdots & & \ddots & \\ -l_{n1} & 0 & & 1 \end{bmatrix}, \quad L_{(1)} = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ l_{21} & 1 & & \\ \vdots & & \ddots & \\ l_{n1} & 0 & & 1 \end{bmatrix} \quad \text{unit lower triangular.}$$

10 Factorise once, substitute twice: $A = LU$

After $n - 1$ steps, $U = L_{(n-1)}^{-1} \cdots L_{(1)}^{-1} A$, so

$$A = LU, \quad L = L_{(1)} \cdots L_{(n-1)}, \quad U = L_{(n-1)}^{-1} \cdots L_{(1)}^{-1} A,$$

with L unit lower triangular (it stores the multipliers) and U upper triangular. The system is then solved by two triangular solves:

$$Ax = f \longrightarrow L(Ux) = f \longrightarrow \underbrace{Ly = f}_{\text{forward substitution}}, \quad \underbrace{Ux = y}_{\text{backward substitution}}.$$

- the substitutions are cheap, $O(n^2)$ for a dense matrix, against $O(n^3)$ for the elimination: **the cost is computing L and U** ;
- the factorisation is done once: many right-hand sides come almost for free;
- for A s.p.d. one can take $U = L^T$ (**Cholesky**), half the work and memory.

What a direct solver costs

	flops	memory
dense $n \times n$	$O(n^3)$	$O(n^2)$
sparse FE matrix, stored	—	$O(n \cdot nnz)$
sparse FE matrix, 3D, factorised (nested dissection)	$O(n^2)$	$O(n^{4/3})$

- FE matrices are **sparse**: nnz non-zeros per row, set by d and the order p (the local DOFs), **not by n** ;
- but elimination creates **fill-in**: L and U are far less sparse than A . Orderings (nested dissection) limit it, up to the bounds above;
- hard to parallelise: not scalable beyond $O(100)$ processors on distributed memory.

In summary: the method of choice for small and medium problems, in 1D and 2D; not viable for large-scale 3D simulations. **There**: **iterative solvers with robust preconditioners**.

11 Iterative solvers: the Richardson method

Approximate u up to a tolerance. The best (multigrid, domain decomposition) reach **linear complexity** and run on millions of processors; too involved for this unit. We study their *building blocks*.

Definition (Preconditioner)

$B = A + E$, E the perturbation. A good B : B^{-1} cheap to apply (and parallel), and $B^{-1}A$ close to I ($\kappa(B^{-1}A) \approx 1$).

$$Au = f \iff Bu = f + Eu. \text{ Iterate: given } u^0, \text{ for } k = 0, 1, \dots$$
$$Bu^{k+1} = f + Eu^k.$$

$B = I$: **Richardson**; otherwise **preconditioned Richardson**. With the residual $r^k \doteq f - Au^k$,

$$Bu^{k+1} = f - Au^k + Bu^k = r^k + Bu^k \implies u^{k+1} = u^k + B^{-1}r^k.$$

If $r^k = 0$ the iteration stops, and $u^k = u$ since A is non-singular. **Relaxation**: take only part of the correction,

$$u^{k+1} = u^k + \alpha B^{-1}(f - Au^k), \quad \alpha > 0 \quad (\text{notes, eq. (6.1)}).$$

12 Error of the Richardson method

$e^k \doteq u - u^k$. Subtract u from (6.1) and use $f = Au$:

$$e^{k+1} = e^k - \alpha B^{-1} A e^k = (I - \alpha B^{-1} A) e^k.$$

A, B s.p.d. $\Rightarrow \lambda_i \doteq \lambda_i(B^{-1}A)$ real positive (Lemma 6.2.1), and the iteration matrix $G \doteq I - \alpha B^{-1}A$ has eigenvalues $1 - \alpha\lambda_i$.

G is not symmetric, so $\|G\|_2 \neq \rho(G)$ in general (it can be > 1 while $\rho < 1$). Measure the error in the B -norm, $\|v\|_B \doteq \sqrt{v^T B v} = \|B^{1/2}v\|_2$:

$$\begin{aligned}\|G\|_B &= \max_{v \neq 0} \frac{\|B^{1/2} G v\|_2}{\|B^{1/2} v\|_2} = \|B^{1/2} G B^{-1/2}\|_2 = \|I - \alpha B^{-1/2} A B^{-1/2}\|_2 \\ &= \max_i |1 - \alpha \lambda_i| = \rho(G),\end{aligned}$$

because $I - \alpha B^{-1/2} A B^{-1/2}$ is symmetric and similar to G . Hence

$$\begin{aligned}\|e^{k+1}\|_B &\leq \rho(G) \|e^k\|_B \leq \rho(G)^{k+1} \|e^0\|_B, \\ \rho(G) &= \max(|1 - \alpha \lambda_{\min}|, |1 - \alpha \lambda_{\max}|).\end{aligned}$$

($B = I$: the B -norm is the Euclidean norm and $G = I - \alpha A$ is symmetric.)

13 Choosing the relaxation parameter

From the previous board, as a function of the relaxation parameter,

$$\rho(\alpha) = \max (|1 - \alpha\lambda_{\min}|, |1 - \alpha\lambda_{\max}|) :$$

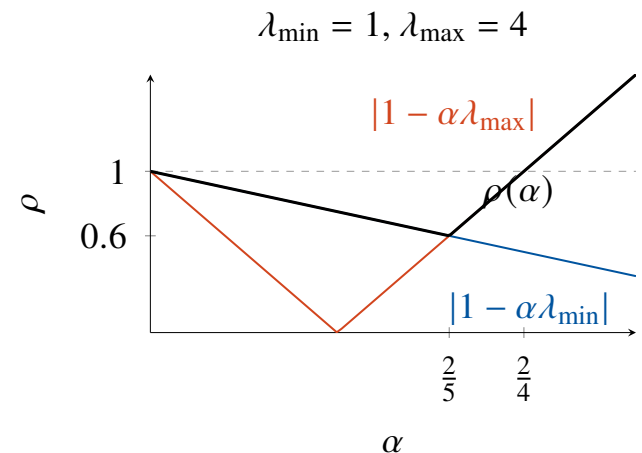
the black curve on the right (blue: $|1 - \alpha\lambda_{\min}|$, orange: $|1 - \alpha\lambda_{\max}|$).

Converges iff $\rho < 1$, i.e. $0 < \alpha < 2/\lambda_{\max}$.

The optimal α balances the two extreme eigenvalues, $1 - \alpha\lambda_{\min} = \alpha\lambda_{\max} - 1$:

$$\alpha_{\text{opt}} = \frac{2}{\lambda_{\max} + \lambda_{\min}},$$

$$\rho_{\text{opt}} = \frac{\lambda_{\max} - \lambda_{\min}}{\lambda_{\max} + \lambda_{\min}} = \frac{\kappa - 1}{\kappa + 1}.$$



But: $\lambda_{\min}, \lambda_{\max}$ cost more than the solve itself; without them the method may not converge; with relaxation it is slow; and there is no bound on the number of iterations. **Better methods are needed.**

14 Steepest descent

For A s.p.d., u is the minimiser of the **energy**

$$\Phi(x) \doteq \frac{1}{2}x^T Ax - f^T x, \quad \text{since} \quad \partial_x \Phi(x) = Ax - f = -r.$$

Richardson ($B = I$) moves along the residual, i.e. along $-\nabla\Phi$: the direction of steepest descent.

Choose α by minimising Φ on that line (prove it!):

$$\alpha_k = \arg \min_{\alpha \in \mathbb{R}} \Phi(u^k + \alpha r^k).$$

Definition (Steepest descent)

Given u^0 , for $k = 0, 1, \dots$:

$$u^{k+1} = u^k + \alpha_k (f - Au^k), \quad \alpha_k = \frac{\langle r^k, r^k \rangle}{\langle r^k, Ar^k \rangle}.$$

15 Steepest descent in pictures

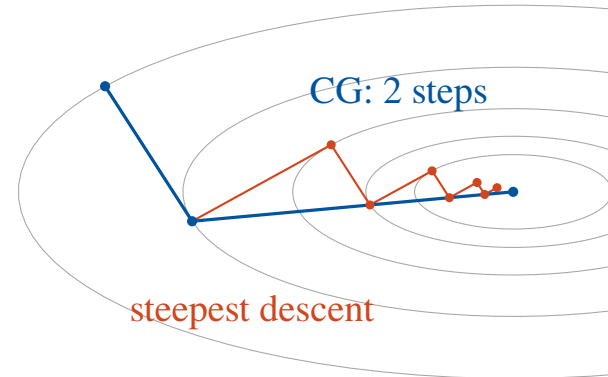
No spectrum needed, and it always converges:

$$\|e^k\|_A \leq \left(\frac{\kappa(A) - 1}{\kappa(A) + 1} \right)^k \|e^0\|_A.$$

Cost per iteration: a sparse matrix–vector product, $O(n)$, plus two inner products.

Iterations grow with $\kappa(A) \sim h^{-2}$: the total cost deteriorates under refinement. The zig-zag on the right is the reason: each step is orthogonal to the previous one.

$A = \text{diag}(1, 6)$, level sets of Φ



16 Conjugate directions

Steepest descent optimises the *distance*, not the *direction*. Suppose we had an *A-orthogonal basis* of conjugate directions $\{p^k\}_{k=0}^{n-1}$,

$$\langle p^i, Ap^j \rangle = 0 \quad (i \neq j), \quad u^{k+1} = u^k + \alpha_k p^k, \quad \alpha_k = \frac{\langle r^k, p^k \rangle}{\langle p^k, Ap^k \rangle}.$$

- each α_k minimises Φ along p^k , and the directions do not interfere: *exact after at most n steps*;
- CG was introduced (1952) as a *direct* solver; rounding errors destroy finite termination;
- rediscovered as an *iterative* method: tiny residual after *far fewer than n* steps. The most used solver in practice for s.p.d. matrices.

17 How to build the conjugate directions

From the residuals, A -orthogonalising against the previous directions (Gram–Schmidt in the A -inner product):

$$p^k = r^k + \sum_{i=0}^{k-1} \beta^i p^i, \quad \beta^i = -\frac{\langle Ar^k, p^i \rangle}{\langle Ap^i, p^i \rangle} \quad \text{so that } \langle Ap^k, p^i \rangle = 0 \quad \forall i < k.$$

Full orthogonalisation stores **all** previous directions and costs more at every step.

Key property of CG (exact arithmetic): the residuals are mutually orthogonal, and $\beta^i = 0$ for $i < k - 1$; the *previous direction only* is enough,

$$p^k = r^k + \beta_{k-1} p^{k-1}.$$

Fixed cost and memory per iteration, whatever k .

18 The conjugate gradient algorithm

Algorithm (Conjugate gradient — notes, Algorithm 6.1)

```
 $p^0 := r^0 := f - Ax^0$   
for  $j = 0, \dots$ , till CONV do  
   $s^{j+1} = Ap^j$   
   $\alpha_j := (r^j, r^j) / (s^{j+1}, p^j)$   
   $x^{j+1} := x^j + \alpha_j p^j$   
   $r^{j+1} := r^j - \alpha_j s^{j+1}$   
   $\beta_j := (r^{j+1}, r^{j+1}) / (r^j, r^j)$   
   $p^{j+1} := r^{j+1} + \beta_j p^j$   
end for
```

Per iteration: one matrix–vector product, two inner products, three updates; three vectors stored.

Only for s.p.d. matrices (Φ must be a quadratic form with a minimum).

Always converges, and $\sqrt{\kappa}$ replaces κ (below): $\sqrt{\kappa(L)} \sim 1/h$, so iterations $\sim 1/h$, not $1/h^2$.

Theorem (Convergence of CG)

$$\|e^k\|_A \leq 2 \left(\frac{\sqrt{\kappa(A)} - 1}{\sqrt{\kappa(A)} + 1} \right)^k \|e^0\|_A.$$

19 Preconditioned conjugate gradient

We want $\kappa \approx 1$. We cannot change A ; we **precondition** with an s.p.d. B and solve instead

$$B^{-\frac{1}{2}}AB^{-\frac{1}{2}}v = B^{-\frac{1}{2}}f, \quad u = B^{-\frac{1}{2}}v.$$

The matrix is still s.p.d., so CG applies, and by Lemma 6.2.1 its condition number is $\kappa(B^{-1}A)$:

$$\|e^k\|_A \leq 2 \left(\frac{\sqrt{\kappa(B^{-1}A)} - 1}{\sqrt{\kappa(B^{-1}A)} + 1} \right)^k \|e^0\|_A.$$

But $B^{-1/2}$ is costly, and the equivalent system $B^{-1}Au = B^{-1}f$ is **not symmetric**.

Rearranging CG (next board): only B^{-1} is ever applied, yet the iterates are exactly those of CG on $B^{-\frac{1}{2}}AB^{-\frac{1}{2}}$.

20 The PCG algorithm

Algorithm (PCG — notes, Algorithm 6.2)

```

$$r^0 := f - Ax^0, z^0 = B^{-1}r^0$$

$$p^0 := z^0$$
for  $j = 0, \dots$ , till CONV do  
     $s^{j+1} = Ap^j$   
     $\alpha_j := (z^j, r^j) / (s^{j+1}, p^j)$   
     $x^{j+1} := x^j + \alpha_j p^j$   
     $r^{j+1} := r^j - \alpha_j s^{j+1}$   
     $z^{j+1} := B^{-1}r^{j+1}$   
     $\beta_j := (z^{j+1}, r^{j+1}) / (z^j, r^j)$   
     $p^{j+1} := z^{j+1} + \beta_j p^j$   
end for
```

Compared with CG: one extra vector z and, per iteration, **one application of B^{-1}** next to the product with A .

$B = I$ gives back CG. $B = \text{diag}(A)$ (**Jacobi**) is the cheapest non-trivial choice.

The open problem: **a cheap, robust B for finite element matrices**, with $\kappa(B^{-1}A)$ bounded independently of h — multigrid, domain decomposition (not in this unit).

21 Summary

	needs	convergence / cost
LU (direct)	any non-singular A	exact; $O(n^3)$ dense, fill-in if sparse
Richardson	$\alpha < 2/\lambda_{\max}$	$\rho = \max_i 1 - \alpha \lambda_i $; needs the spectrum
Steepest descent	A s.p.d.	$\left(\frac{\kappa-1}{\kappa+1}\right)^k$; $O(n)$ per iteration
CG	A s.p.d.	$2\left(\frac{\sqrt{\kappa}-1}{\sqrt{\kappa}+1}\right)^k$; $\leq n$ steps exactly
PCG	A, B s.p.d.	same with $\kappa(B^{-1}A)$; one B^{-1} per iteration

- **Dictionary:** $u^T M u = \|u_h\|_{L^2}^2$, $u^T L u = |u_h|_{H^1}^2$; $\kappa(M) = O(1)$, $\kappa(L) = O(h^{-2})$;
- iterations follow κ (or $\sqrt{\kappa}$): **refinement makes the solve harder**;
- a preconditioner replaces $\kappa(A)$ by $\kappa(B^{-1}A)$: **that is where the research is**.